Storage backend

Astra Automation

NetApp
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목차

Storage backend 1
Storage backend overview 1
Create a new storage backend 1
List all storage backends 17
Retrieve a storage backend 31
Modify a storage backend 45
Delete a storage backend 57

Storage backend

Storage backend overview

The Storage Backend API provides access to representations of providers of storage services (storage backends) that can be used by managed clusters and applications.

Create a new storage backend

POST `/accounts/{account_id}/topology/v1/storageBackends`

Indicates desired values for the Storage Backend API resource to be created.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}

Request Body

Indicates desired values for the Storage Backend API resource to be created.

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-storageBackend"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2" • "1.3"

Name	Type	Required	Description
backendName	string	False	JSON string containing a user-specified name for the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
backendType	string	True	JSON string containing a value indicating the type of the storage backend being represented. Defined values are: • "ontap" - The storage backend represents a NetApp ONTAP system • enum: ["ontap"] • example: ontap
backendVersion	string	False	JSON string containing a value indicating the version of the storage backend being represented. Defined values are: • Minimum length: 1 • Maximum length: 63
backendCredentialsName	string	False	JSON string containing a value indicating the name of one or more credential resources used for the storage backend. Allows for credential rotation where multiple credential objects have the same name. Defined values are: • Minimum length: 1 • Maximum length: 63

Name	Type	Required	Description
metadata	type_astra_metadata_update	False	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example request

```
{
  "type": "application/astra-storageBackend",
  "version": "1.3",
  "backendName": "st1-45",
  "backendType": "ontap",
  "backendCredentialsName": "st1-45-cred"
}
```

Response

```
Status: 201, Returns the newly created storageBackend resource in the JSON
response body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-storageBackend"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2" • "1.3"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
backendName	string	True	JSON string containing a user-specified name for the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
backendType	string	True	JSON string containing a value indicating the type of the storage backend being represented. Defined values are: • "ontap" - The storage backend represents a NetApp ONTAP system • enum: ["ontap"] • example: ontap
backendVersion	string	True	JSON string containing a value indicating the version of the storage backend being represented. Defined values are: • Minimum length: 1 • Maximum length: 63

Name	Type	Required	Description
backendCredentialsName	string	True	JSON string containing a value indicating the name of one or more credential resources used for the storage backend. Allows for credential rotation where multiple credential objects have the same name. Defined values are: • Minimum length: 1 • Maximum length: 63
configVersion	string	False	JSON string containing a value indicating the active configuration to be used by the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
state	string	True	JSON string containing a value indicating the operational state of the storage backend. Defined values are: • "discovered" - The storage backend has been discovered from a managed cluster • "running" - The storage backend is running normally • "unknown" - The state of the storage backend cannot be determined • "failed" - The storage backend is in an unavailable state

Name	Type	Required	Description
stateDesired	string	False	JSON string containing a value indicating the desired operational state of the storage backend. Defined values are: • "running" - The storage backend will be in a normal running state
stateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a storage backend is unable to transition to "running", why it has transitioned to "failed" or "unknown", and/or why it is at risk of transitioning to "failed". Defined values are: • Minimum length: 1 • Maximum length: 127
managedState	string	True	JSON string containing a value indicating the management state of the storage backend. Defined values are: • "pending" - The storage backend is scheduled to be discovered or added • "unmanaged" - The storage backend is in a "discovered" state, is not under management, and is eligible for management • "managed" - The storage backend is in a "running" or "unknown" state and is under management

Name	Type	Required	Description
managedStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why the storage backend is ineligible to be managed. Defined values are: • Minimum length: 1 • Maximum length: 127
healthState	string	False	JSON string containing a value indicating the health state of the storage backend. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a storage backend is not in a normal health state. Defined values are: • Minimum length: 1 • Maximum length: 127
protectionState	string	True	JSON string containing a value indicating the protection state of the storage backend. "unknown" for ONTAP. Defined values are: • "protected" - The storage backend has a protection policy defined and is in conformance with that policy • "partial" - The storage backend has a protection policy defined but is out of conformance with that policy • "none" - The storage backend is not protected • "unknown" - The storage backend protection policy conformance cannot be determined • enum: ["protected", "partial", "none", "unknown"] • example: protected

Name	Type	Required	Description
protectionStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a protection policy is out of conformance. Defined values are: • Minimum length: 1 • Maximum length: 127
capabilities	capabilities	True	JSON object containing this Storage Backend capabilities.
ontap	ontap	False	JSON object containing specific information related to ONTAP clusters. • additionalProperties: false
metadata	type_astra_metadata_update	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example response


```
{
  "type": "application/astra-storageBackend",
  "version": "1.3",
  "id": "04ad4237-3c89-5ca5-8a40-282ab5eb6428",
  "backendName": "st1-45",
  "backendType": "ontap",
  "backendVersion": "9.8",
  "backendCredentialsName": "st1-45-cred",
  "state": "running",
  "stateUnready": [],
  "managedState": "managed",
  "managedStateUnready": [],
  "protectionState": "unknown",
  "protectionStateUnready": [
    "Waiting for storage backend discovery"
  ],
  "ontap": {
    "authenticationStyle": "basic",
    "backendManagementIP": "10.193.179.105",
    "managementIPs": [
      "10.193.188.110",
      "10.193.179.105",
      "10.193.179.106"
    ]
  },
  "capabilities": {
    "flexClone": "true",
    "snapMirror": "true",
    "s3": "true"
  },
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 409, Conflict
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/10",
  "title": "JSON resource conflict",
  "detail": "The request body JSON contains a field that conflicts with
an idempotent value.",
  "status": "409"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

capabilities

JSON object containing this Storage Backend capabilities.

Name	Type	Required	Description
flexClone	string	True	JSON string indicating if this Storage Backend supports FlexClone.
snapMirror	string	True	JSON string indicating if this Storage Backend supports SnapMirror.
s3	string	True	JSON string indicating if this Storage Backend supports s3.

ontap

JSON object containing specific information related to ONTAP clusters.

Name	Type	Required	Description
authenticationStyle	string	True	JSON string indicating whether the ONTAP authentication is performed using a username/password combination or a certificate. Defined values are: • "basic" - Uses username/password • "certificate" - Uses certificate • enum: ["basic", "certificate"] • example: basic
backendManagementIP	string	False	JSON string containing the IP address used to connect to this ONTAP cluster.
managementIPs	array[string]	False	JSON array of JSON strings containing the list of all possible management IPs used to connect to this ONTAP cluster. This is used to correlate backends detected through Trident and storage backends in Astra. • uniqueItems: 1

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

invalidFields

Name	Type	Required	Description
name	string	True	Name of the invalid request body field
reason	string	True	Reason why the request body field is invalid

List all storage backends

GET `/accounts/{account_id}/topology/v1/storageBackends`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,name,state

Name	Type	In	Required	Description
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: storageBackend_1.3_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response


```
{
  "type": "application/astra-storageBackends",
  "version": "1.3",
  "items": [
    [
      "d6ee1e51-d65f-415a-a3d4-6d7fa8b64943",
      "st1-45",
      "running"
    ],
    [
      "a5c508b7-aacd-4d19-9455-53c0d183ab4e",
      "st2-67",
      "discovered"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

capabilities

JSON object containing this Storage Backend capabilities.

Name	Type	Required	Description
flexClone	string	True	JSON string indicating if this Storage Backend supports FlexClone.
snapMirror	string	True	JSON string indicating if this Storage Backend supports SnapMirror.
s3	string	True	JSON string indicating if this Storage Backend supports s3.

ontap

JSON object containing specific information related to ONTAP clusters.

Name	Type	Required	Description
authenticationStyle	string	True	JSON string indicating whether the ONTAP authentication is performed using a username/password combination or a certificate. Defined values are: • "basic" - Uses username/password • "certificate" - Uses certificate • enum: ["basic", "certificate"] • example: basic
backendManagementIP	string	False	JSON string containing the IP address used to connect to this ONTAP cluster.

Name	Type	Required	Description
managementIPs	array[string]	False	JSON array of JSON strings containing the list of all possible management IPs used to connect to this ONTAP cluster. This is used to correlate backends detected through Trident and storage backends in Astra. • uniqueItems: 1

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

storageBackend_1.3_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-storageBackend"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2" • "1.3"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
backendName	string	True	JSON string containing a user-specified name for the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
backendType	string	True	JSON string containing a value indicating the type of the storage backend being represented. Defined values are: • "ontap" - The storage backend represents a NetApp ONTAP system • enum: ["ontap"] • example: ontap
backendVersion	string	True	JSON string containing a value indicating the version of the storage backend being represented. Defined values are: • Minimum length: 1 • Maximum length: 63

Name	Type	Required	Description
backendCredentialsName	string	True	JSON string containing a value indicating the name of one or more credential resources used for the storage backend. Allows for credential rotation where multiple credential objects have the same name. Defined values are: • Minimum length: 1 • Maximum length: 63
configVersion	string	False	JSON string containing a value indicating the active configuration to be used by the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
state	string	True	JSON string containing a value indicating the operational state of the storage backend. Defined values are: • "discovered" - The storage backend has been discovered from a managed cluster • "running" - The storage backend is running normally • "unknown" - The state of the storage backend cannot be determined • "failed" - The storage backend is in an unavailable state

Name	Type	Required	Description
stateDesired	string	False	JSON string containing a value indicating the desired operational state of the storage backend. Defined values are: • "running" - The storage backend will be in a normal running state
stateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a storage backend is unable to transition to "running", why it has transitioned to "failed" or "unknown", and/or why it is at risk of transitioning to "failed". Defined values are: • Minimum length: 1 • Maximum length: 127
managedState	string	True	JSON string containing a value indicating the management state of the storage backend. Defined values are: • "pending" - The storage backend is scheduled to be discovered or added • "unmanaged" - The storage backend is in a "discovered" state, is not under management, and is eligible for management • "managed" - The storage backend is in a "running" or "unknown" state and is under management

Name	Type	Required	Description
managedStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why the storage backend is ineligible to be managed. Defined values are: • Minimum length: 1 • Maximum length: 127
healthState	string	False	JSON string containing a value indicating the health state of the storage backend. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a storage backend is not in a normal health state. Defined values are: • Minimum length: 1 • Maximum length: 127
protectionState	string	True	JSON string containing a value indicating the protection state of the storage backend. "unknown" for ONTAP. Defined values are: • "protected" - The storage backend has a protection policy defined and is in conformance with that policy • "partial" - The storage backend has a protection policy defined but is out of conformance with that policy • "none" - The storage backend is not protected • "unknown" - The storage backend protection policy conformance cannot be determined • enum: ["protected", "partial", "none", "unknown"] • example: protected

Name	Type	Required	Description
protectionStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a protection policy is out of conformance. Defined values are: • Minimum length: 1 • Maximum length: 127
capabilities	capabilities	True	JSON object containing this Storage Backend capabilities.
ontap	ontap	False	JSON object containing specific information related to ONTAP clusters. • additionalProperties: false
metadata	type_astra_metadata_update	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	

Name	Type	Required	Description
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve a storage backend

GET `/accounts/{account_id}/topology/v1/storageBackends/{storageBackend_id}`

Returns the stored Storage Backend API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
storageBackend_id	string	path	True	ID of the storageBackend collection to list

Response

```
Status: 200, Returns the stored Storage Backend API resource in the JSON
response body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-storageBackend"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2" • "1.3"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
backendName	string	True	JSON string containing a user-specified name for the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
backendType	string	True	JSON string containing a value indicating the type of the storage backend being represented. Defined values are: • "ontap" - The storage backend represents a NetApp ONTAP system • enum: ["ontap"] • example: ontap

Name	Type	Required	Description
backendVersion	string	True	JSON string containing a value indicating the version of the storage backend being represented. Defined values are: • Minimum length: 1 • Maximum length: 63
backendCredentialsName	string	True	JSON string containing a value indicating the name of one or more credential resources used for the storage backend. Allows for credential rotation where multiple credential objects have the same name. Defined values are: • Minimum length: 1 • Maximum length: 63
configVersion	string	False	JSON string containing a value indicating the active configuration to be used by the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63

Name	Type	Required	Description
state	string	True	JSON string containing a value indicating the operational state of the storage backend. Defined values are: • "discovered" - The storage backend has been discovered from a managed cluster • "running" - The storage backend is running normally • "unknown" - The state of the storage backend cannot be determined • "failed" - The storage backend is in an unavailable state
stateDesired	string	False	JSON string containing a value indicating the desired operational state of the storage backend. Defined values are: • "running" - The storage backend will be in a normal running state
stateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a storage backend is unable to transition to "running", why it has transitioned to "failed" or "unknown", and/or why it is at risk of transitioning to "failed". Defined values are: • Minimum length: 1 • Maximum length: 127

Name	Type	Required	Description
managedState	string	True	JSON string containing a value indicating the management state of the storage backend. Defined values are: • "pending" - The storage backend is scheduled to be discovered or added • "unmanaged" - The storage backend is in a "discovered" state, is not under management, and is eligible for management • "managed" - The storage backend is in a "running" or "unknown" state and is under management
managedStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why the storage backend is ineligible to be managed. Defined values are: • Minimum length: 1 • Maximum length: 127

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the storage backend. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a storage backend is not in a normal health state. Defined values are: • Minimum length: 1 • Maximum length: 127

Name	Type	Required	Description
protectionState	string	True	JSON string containing a value indicating the protection state of the storage backend. "unknown" for ONTAP. Defined values are: • "protected" - The storage backend has a protection policy defined and is in conformance with that policy • "partial" - The storage backend has a protection policy defined but is out of conformance with that policy • "none" - The storage backend is not protected • "unknown" - The storage backend protection policy conformance cannot be determined • enum: ["protected", "partial", "none", "unknown"] • example: protected
protectionStateUnready	array[string]	True	JSON array of JSON strings, each indicating a reason why a protection policy is out of conformance. Defined values are: • Minimum length: 1 • Maximum length: 127
capabilities	capabilities	True	JSON object containing this Storage Backend capabilities.

Name	Type	Required	Description
ontap	ontap	False	JSON object containing specific information related to ONTAP clusters. • additionalProperties: false
metadata	type_astra_metadata_update	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example response


```
{
  "type": "application/astra-storageBackend",
  "version": "1.3",
  "id": "04ad4237-3c89-5ca5-8a40-282ab5eb6428",
  "backendName": "st1-45",
  "backendType": "ontap",
  "backendVersion": "9.8",
  "backendCredentialsName": "st1-45-cred",
  "state": "discovered",
  "stateUnready": [],
  "managedState": "unmanaged",
  "managedStateUnready": [],
  "protectionState": "unknown",
  "protectionStateUnready": [
    "Waiting for storage backend discovery"
  ],
  "capabilities": {
    "flexClone": "true",
    "snapMirror": "true",
    "s3": "true"
  },
  "ontap": {
    "authenticationStyle": "basic",
    "backendManagementIP": "10.193.179.105",
    "managementIPs": [
      "10.193.188.110",
      "10.193.179.105",
      "10.193.179.106"
    ]
  },
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

capabilities

JSON object containing this Storage Backend capabilities.

Name	Type	Required	Description
flexClone	string	True	JSON string indicating if this Storage Backend supports FlexClone.
snapMirror	string	True	JSON string indicating if this Storage Backend supports SnapMirror.
s3	string	True	JSON string indicating if this Storage Backend supports s3.

ontap

JSON object containing specific information related to ONTAP clusters.

Name	Type	Required	Description
authenticationStyle	string	True	JSON string indicating whether the ONTAP authentication is performed using a username/password combination or a certificate. Defined values are: • "basic" - Uses username/password • "certificate" - Uses certificate • enum: ["basic", "certificate"] • example: basic
backendManagementIP	string	False	JSON string containing the IP address used to connect to this ONTAP cluster.

Name	Type	Required	Description
managementIPs	array[string]	False	JSON array of JSON strings containing the list of all possible management IPs used to connect to this ONTAP cluster. This is used to correlate backends detected through Trident and storage backends in Astra. • uniqueItems: 1

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Modify a storage backend

PUT `/accounts/{account_id}/topology/v1/storageBackends/{storageBackend_id}`

Replaces the entire stored resource with the supplied JSON request body while preserving key values that are not user-modifiable.

Side Effects

- Specifying a configuration version results in storage backend type-specific configuration changes being applied, which can include the initial installation of a storage backend.
- Specifying a configuration version might result in storageClass resources being added, modified, or removed.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
storageBackend_id	string	path	True	ID of the storageBackend collection to list

Request Body

Replaces the entire stored resource with the supplied JSON request body while preserving key values that are not user-modifiable.

Side Effects

- Specifying a configuration version results in storage backend type-specific configuration changes being applied, which can include the initial installation of a storage backend.
- Specifying a configuration version might result in storageClass resources being added, modified, or removed.

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-storageBackend"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2" • "1.3"
id	string	False	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
backendName	string	False	JSON string containing a user-specified name for the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
backendType	string	False	JSON string containing a value indicating the type of the storage backend being represented. Defined values are: • "ontap" - The storage backend represents a NetApp ONTAP system • enum: ["ontap"] • example: ontap
backendVersion	string	False	JSON string containing a value indicating the version of the storage backend being represented. Defined values are: • Minimum length: 1 • Maximum length: 63

Name	Type	Required	Description
backendCredentialsName	string	False	JSON string containing a value indicating the name of one or more credential resources used for the storage backend. Allows for credential rotation where multiple credential objects have the same name. Defined values are: • Minimum length: 1 • Maximum length: 63
configVersion	string	False	JSON string containing a value indicating the active configuration to be used by the storage backend. Defined values are: • Minimum length: 1 • Maximum length: 63
state	string	False	JSON string containing a value indicating the operational state of the storage backend. Defined values are: • "discovered" - The storage backend has been discovered from a managed cluster • "running" - The storage backend is running normally • "unknown" - The state of the storage backend cannot be determined • "failed" - The storage backend is in an unavailable state

Name	Type	Required	Description
stateDesired	string	False	JSON string containing a value indicating the desired operational state of the storage backend. Defined values are: • "running" - The storage backend will be in a normal running state
stateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a storage backend is unable to transition to "running", why it has transitioned to "failed" or "unknown", and/or why it is at risk of transitioning to "failed". Defined values are: • Minimum length: 1 • Maximum length: 127
managedState	string	False	JSON string containing a value indicating the management state of the storage backend. Defined values are: • "pending" - The storage backend is scheduled to be discovered or added • "unmanaged" - The storage backend is in a "discovered" state, is not under management, and is eligible for management • "managed" - The storage backend is in a "running" or "unknown" state and is under management

Name	Type	Required	Description
managedStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why the storage backend is ineligible to be managed. Defined values are: • Minimum length: 1 • Maximum length: 127
healthState	string	False	JSON string containing a value indicating the health state of the storage backend. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a storage backend is not in a normal health state. Defined values are: • Minimum length: 1 • Maximum length: 127
protectionState	string	False	JSON string containing a value indicating the protection state of the storage backend. "unknown" for ONTAP. Defined values are: • "protected" - The storage backend has a protection policy defined and is in conformance with that policy • "partial" - The storage backend has a protection policy defined but is out of conformance with that policy • "none" - The storage backend is not protected • "unknown" - The storage backend protection policy conformance cannot be determined • enum: ["protected", "partial", "none", "unknown"] • example: protected

Name	Type	Required	Description
protectionStateUnready	array[string]	False	JSON array of JSON strings, each indicating a reason why a protection policy is out of conformance. Defined values are: • Minimum length: 1 • Maximum length: 127
ontap	ontap	False	JSON object containing specific information related to ONTAP clusters. • additionalProperties: false
metadata	type_astra_metadata_update	False	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example request

```
{
  "type": "application/astra-storageBackend",
  "version": "1.3",
  "backendName": "st1-46"
}
```

Response

```
Status: 204, Indicates if the storageBackend resource was updated.
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 409, Conflict
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/10",
  "title": "JSON resource conflict",
  "detail": "The request body JSON contains a field that conflicts with
an idempotent value.",
  "status": "409"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/1",
  "title": "Resource not found",
  "detail": "The resource specified in the request URI wasn't found.",
  "status": "404"
}
```

Definitions

See Definitions

ontap

JSON object containing specific information related to ONTAP clusters.

Name	Type	Required	Description
backendManagementIP	string	False	JSON string containing the IP address used to connect to this ONTAP cluster.
managementIPs	array[string]	False	JSON array of JSON strings containing the list of all possible management IPs used to connect to this ONTAP cluster. This is used to correlate backends detected through Trident and storage backends in Astra. • uniqueItems: 1

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

invalidFields

Name	Type	Required	Description
name	string	True	Name of the invalid request body field
reason	string	True	Reason why the request body field is invalid

Delete a storage backend

DELETE `/accounts/{account_id}/topology/v1/storageBackends/{storageBackend_id}`

Removes the specified Storage Backend API resource.

Side Effects

- All backendConfig resources associated with the storage backend are removed.
- All backendNode resources associated with the storage backend are removed.
- The storage backend might be removed.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
storageBackend_id	string	path	True	ID of the storageBackend collection to list

Response

```
Status: 204, Indicates if the storageBackend resource was deleted.
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/1",
  "title": "Resource not found",
  "detail": "The resource specified in the request URI wasn't found.",
  "status": "404"
}
```

Definitions

See Definitions

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

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